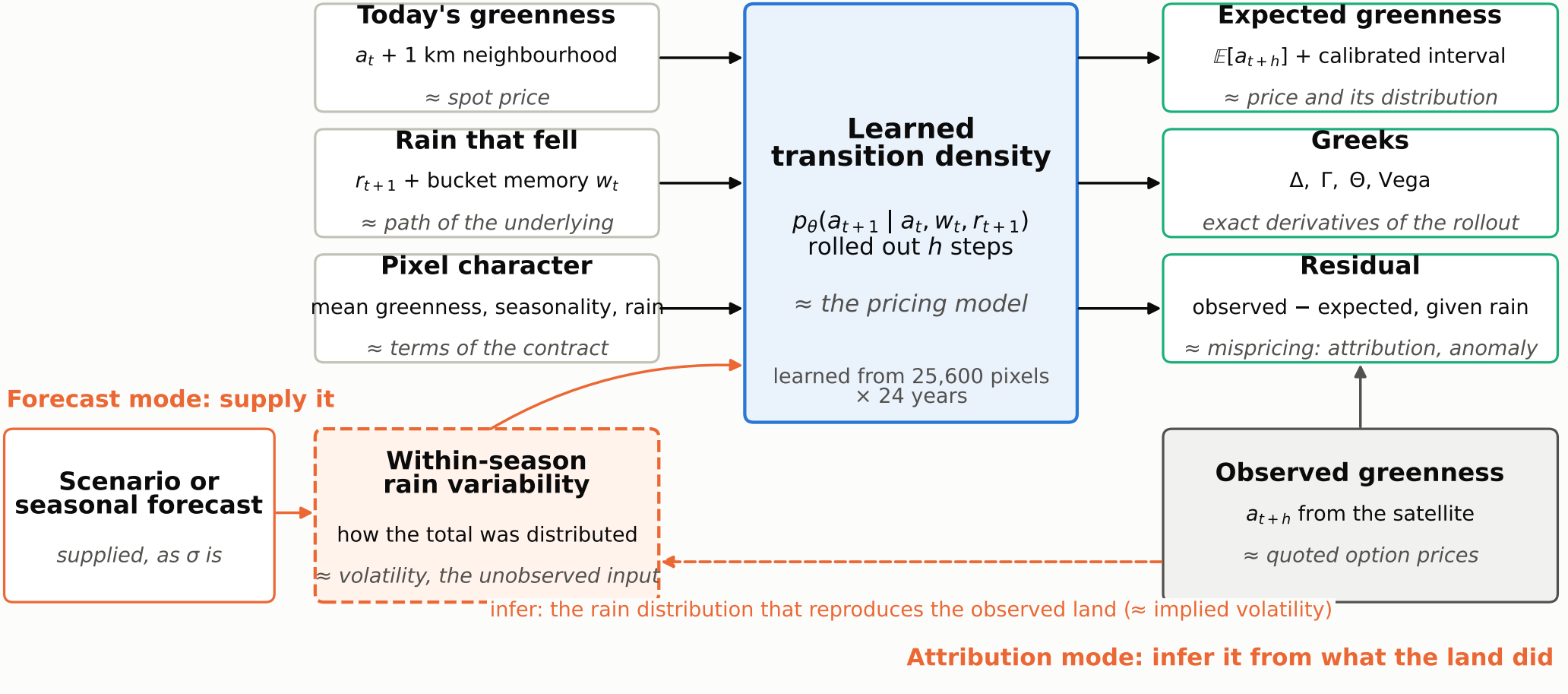
Today's greenness
a_t + 1 km neighbourhood
≈ spot price
Rain that fell
r_{t+1} + bucket memory w_t
≈ path of the underlying
Pixel character
mean greenness, seasonality, rain
≈ terms of the contract
Learned transition density
$p_\theta(a_{t+1} \mid a_t, w_t, r_{t+1})$
rolled out h steps
≈ the pricing model
learned from 25,600 pixels × 24 years
Expected greenness
$\mathbb{E}[a_{t+h}]$ + calibrated interval
≈ price and its distribution
Greeks
Δ, Γ, Θ, Vega
exact derivatives of the rollout
Residual
observed − expected, given rain
≈ mispricing: attribution, anomaly
Forecast mode: supply it
Scenario or seasonal forecast
supplied, as σ is
Within-season rain variability
how the total was distributed
≈ volatility, the unobserved input
Observed greenness
a_{t+h} from the satellite
≈ quoted option prices
infer: the rain distribution that reproduces the observed land (≈ implied volatility)
Attribution mode: infer it from what the land did